



Vizuara AL Labs Presents

Machine Learning in Production

Lecture 3 Notes

Before you model: Planning and scoping a project

What will we cover in this lecture?

- 1) Defining effective planning strategies for ML project work
- 2) Using efficient methods to evaluate potential solutions to an ML problem



RECAP

In the last lecture we have covered

- 1) Difference between data scientist and machine learning engineer
- 2) Focusing on simplicity in all project work to reduce risk
- 3) applying agile fundamentals to ml project work
- 4) differences and similarities between DevOps and MLOps



What Really Fails ML Projects?

- ❌ Not algorithms
- ❌ Not data
- ❌ Not the tools/platform
- ✅ It's poor planning & scoping!

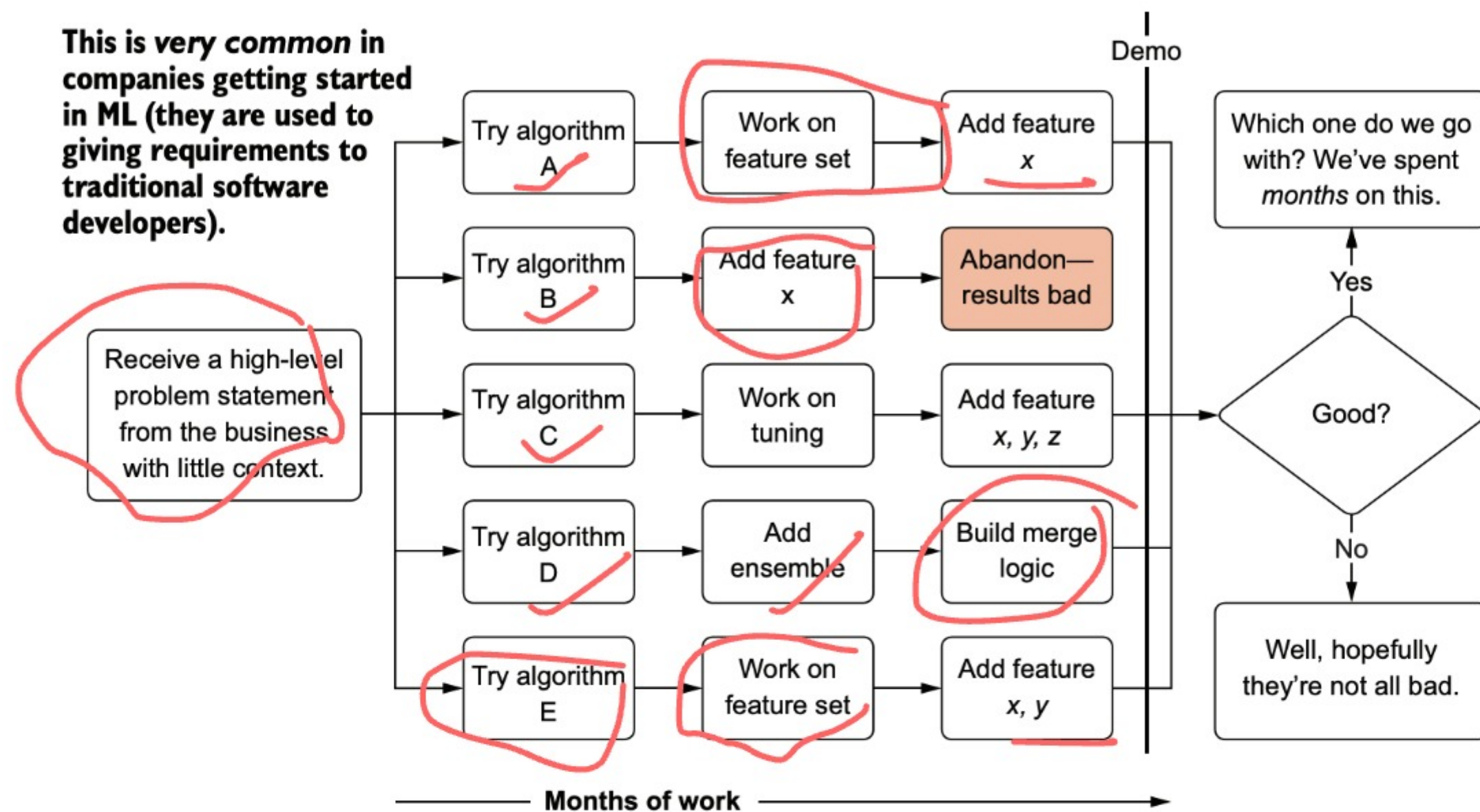
Planning and scoping are critical in ML Engineering because they align technical efforts with real business needs, ensuring that models solve the right problems. Without this foundation, even the most accurate models risk becoming unused solutions that fail to deliver value in production.

The Paradigm Shift in ML

- 1) From "How do we build it?" → To "What are we solving and for whom?"
- 2) Cross-functional planning = better outcomes
- 3) Reduced experimentation time ⌚
- 4) More business-ready solutions ✅

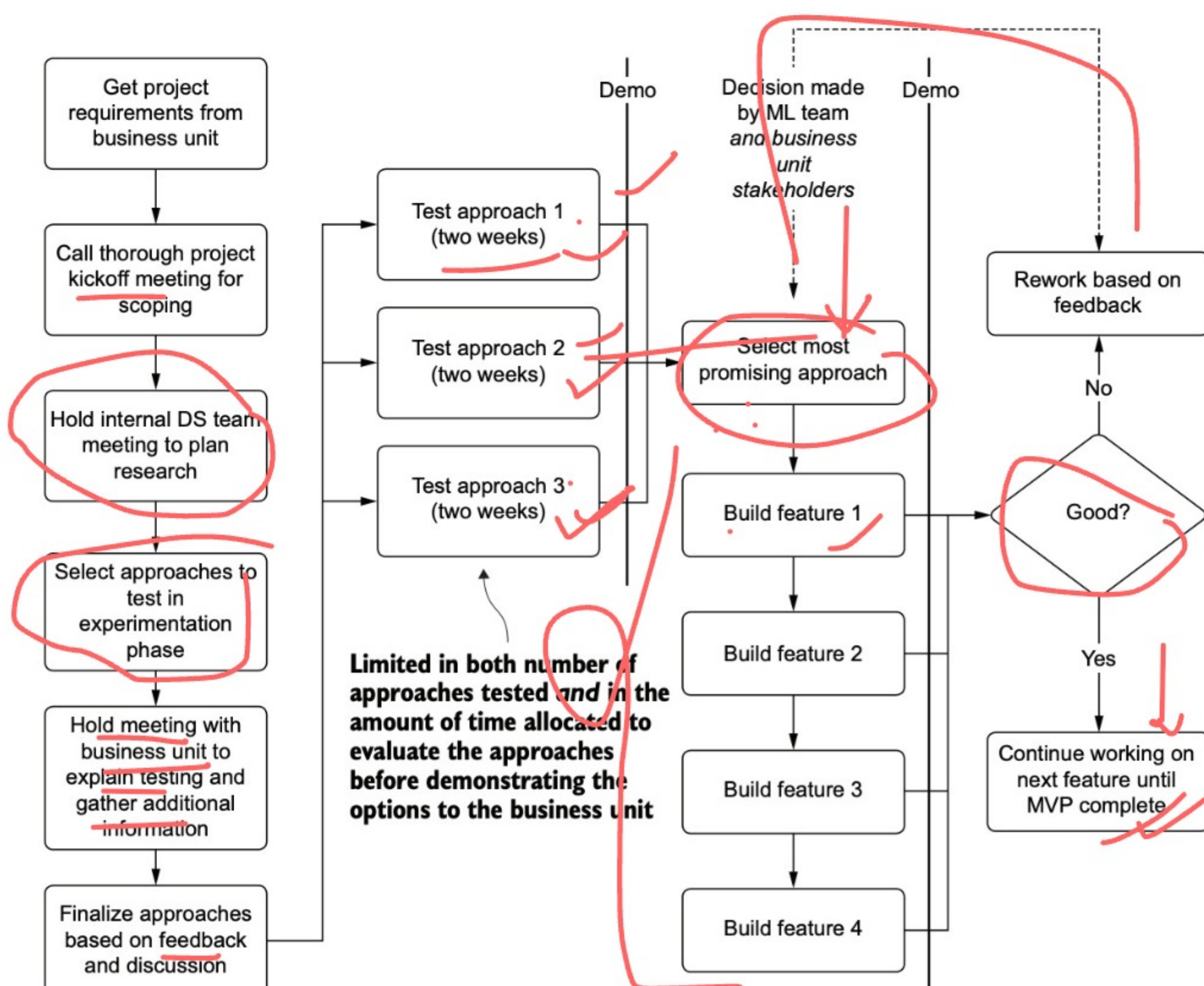
Let us consider 2 Scenarios

Scenario 1: Inadequate Planning and Scoping



Source:- Machine learning in action, author:- Ben Wilson

Scenario 2: Adequate Planning and Scoping



Source:- Machine learning in action, author:- Ben Wilson

3.1) Planning: You want me to predict what?!

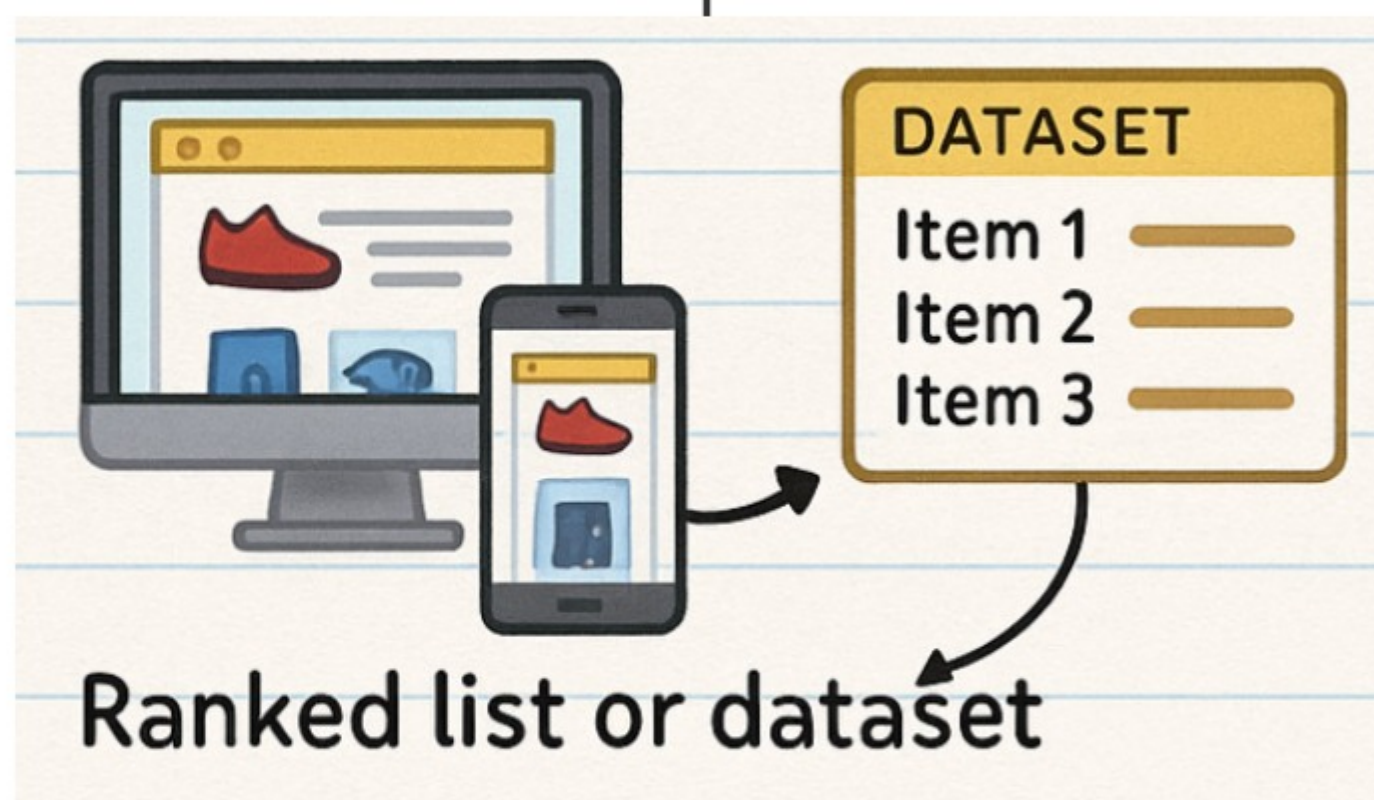
Let's Understand using Case Study



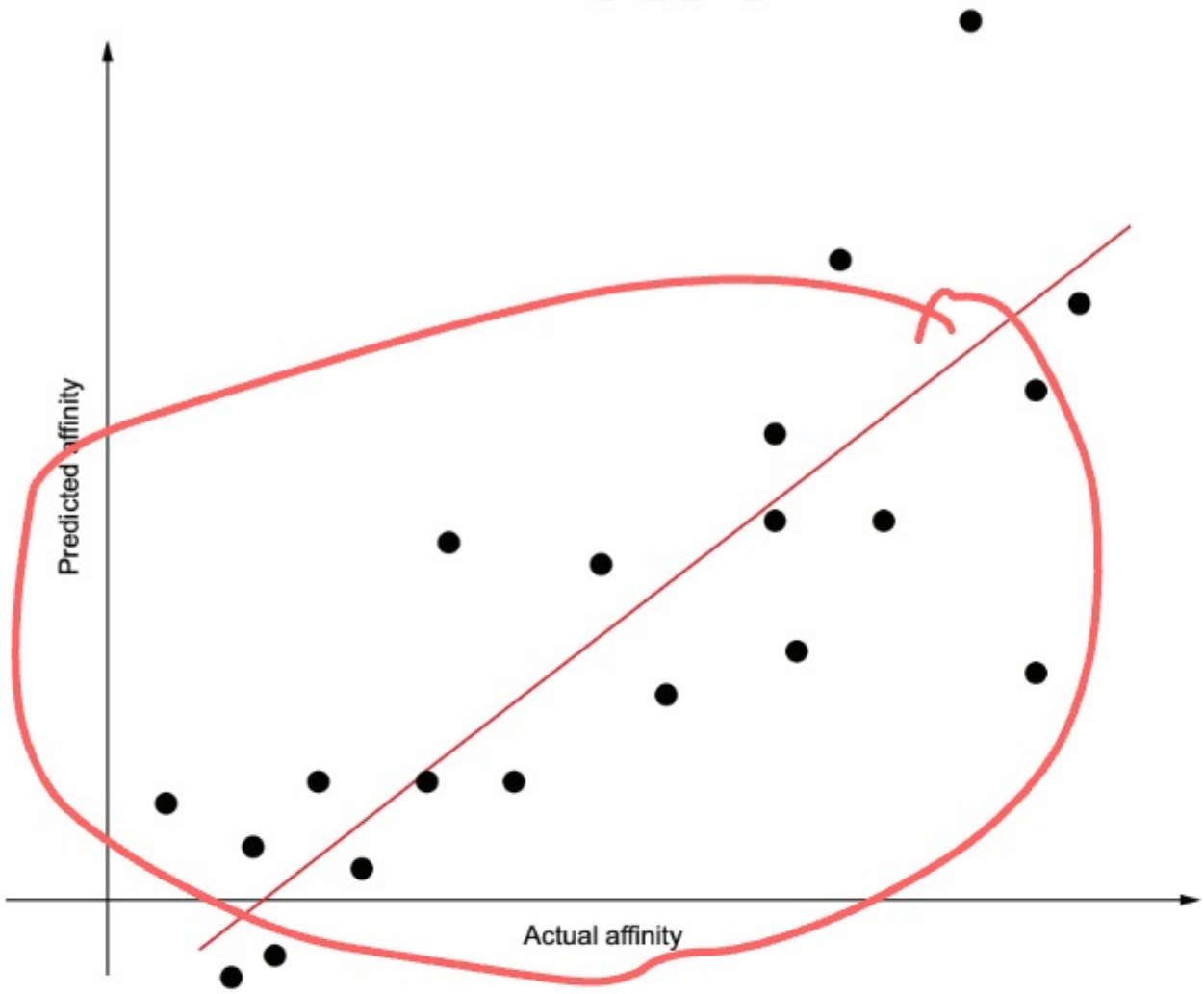
E-Commerce Company ✓✓



BS want personalized recommendation for each user



ML Team starts creating dataset and finds relevant algorithm for the company requirements



ML Team Shows the model working using such graphs and evaluate them using RMSE...BS Team Little Confuse



User ID	Item ID	Actual ranking	Predicted ranking	DCG value
38873	25	1	3	3.0
38873	17	2	2	1.26186
38873	23	3	6	3.0
38873	11	4	1	0.403677
38873	19	5	5	1.934264
38873	3	6	4	1.424829

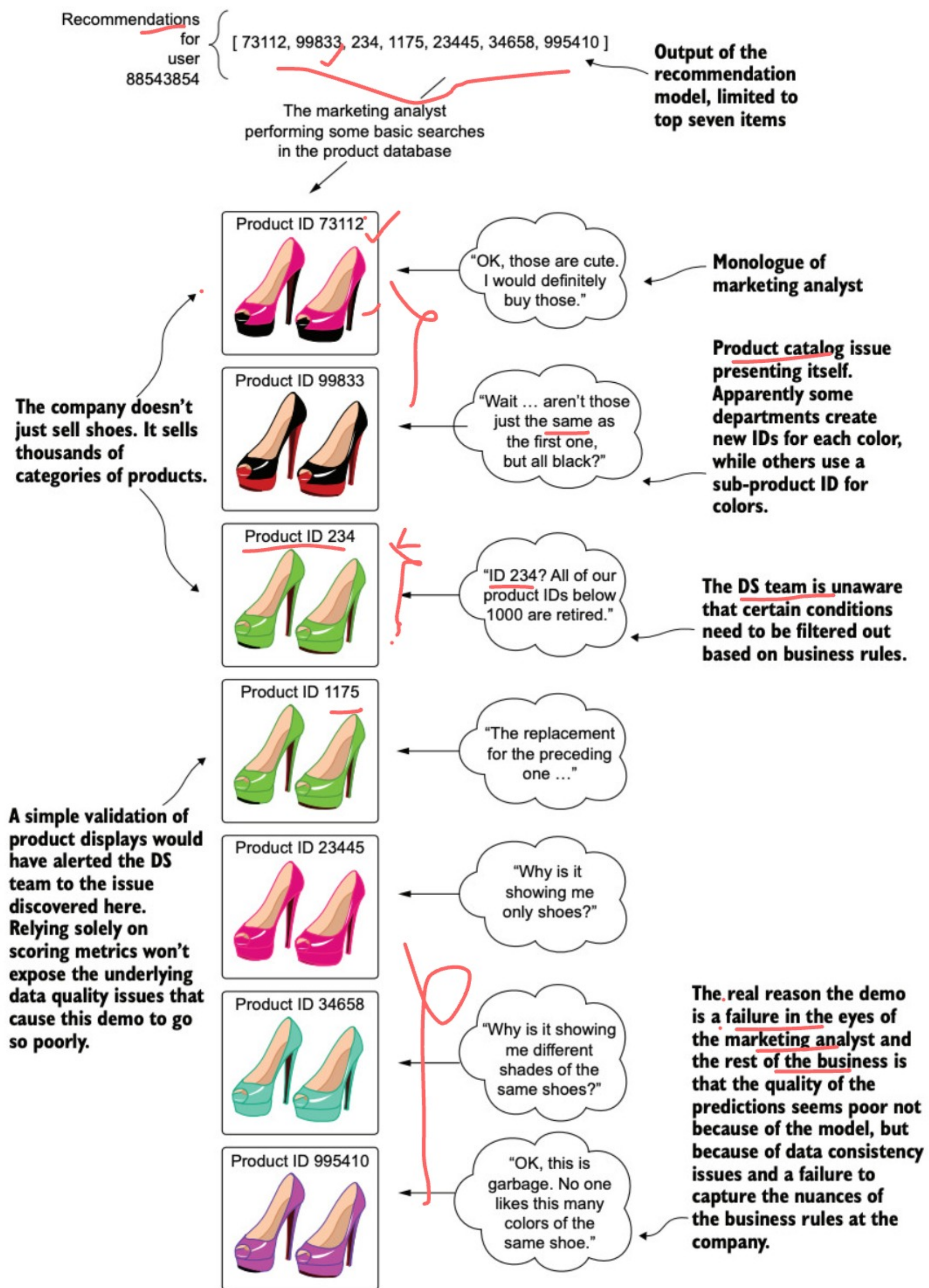
NDCG to illustrate the predictive power of a single user chosen at random

NDCG?



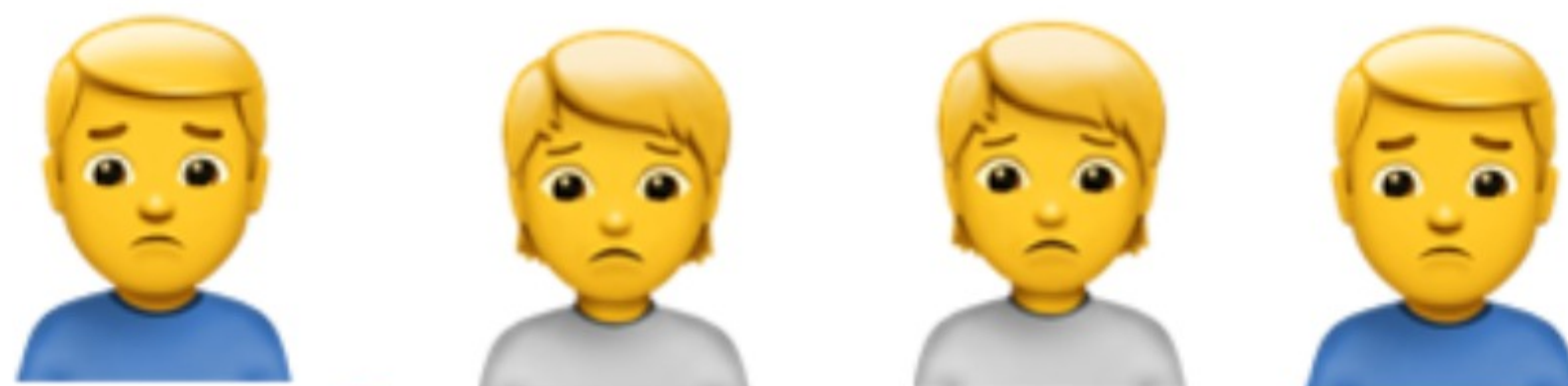
More confusion Created





Source:- Machine learning in action, author:- Ben Wilson

Market Analyst found lot of issues in the system while doing experiment on sample dataset



Entire team is Unhappy

The DS team failed to align with domain experts who had deep knowledge of the data and product. This wasn't an individual failure but a collective oversight in scoping and clarifying the project's real business needs.

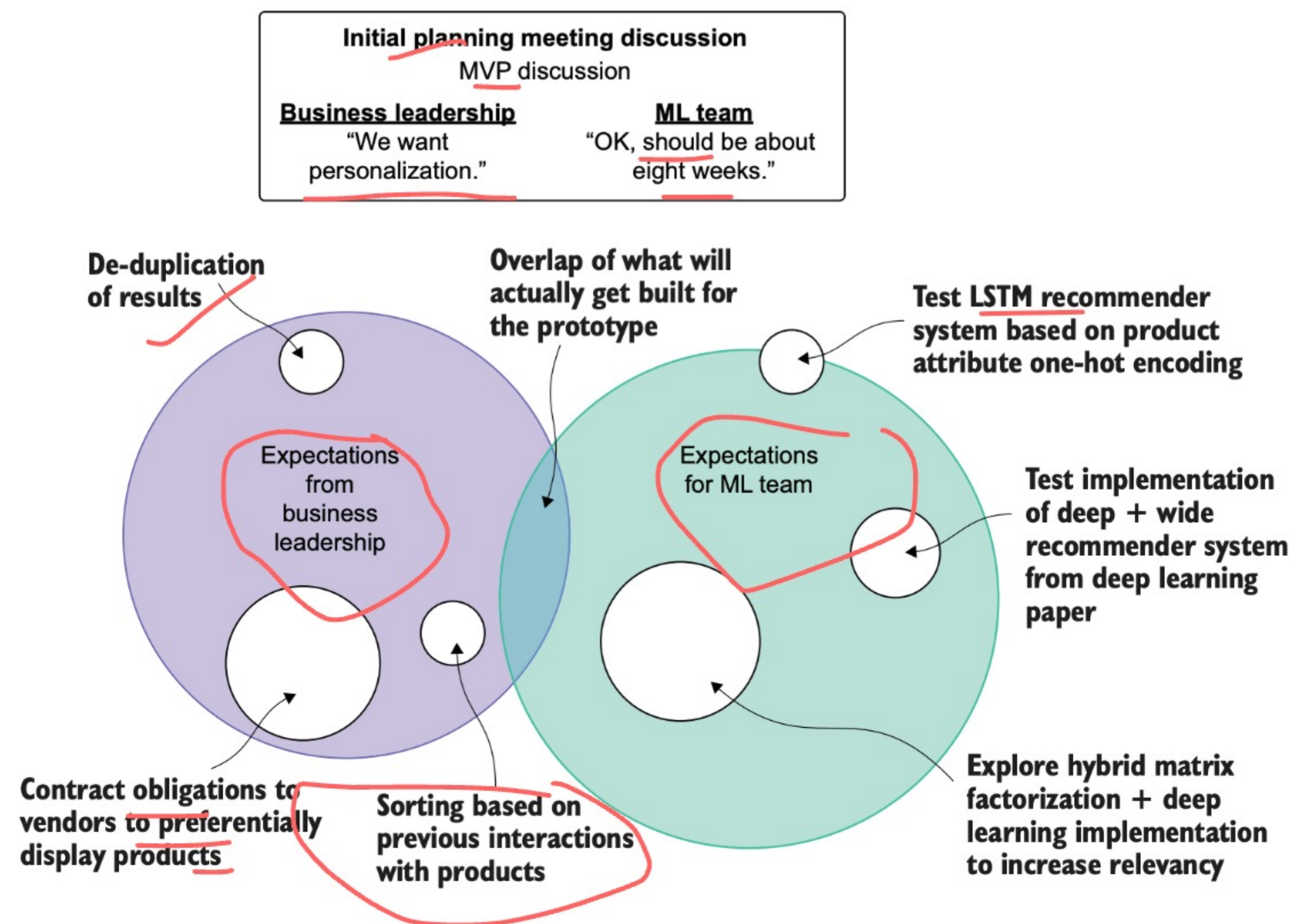
3.2) Basic planning for a project

Before making any assumptions about the project, it is important for both the Data Science team and business stakeholders to begin with a discussion. The conversation should start with one critical question: **"Why are we building this?"**?

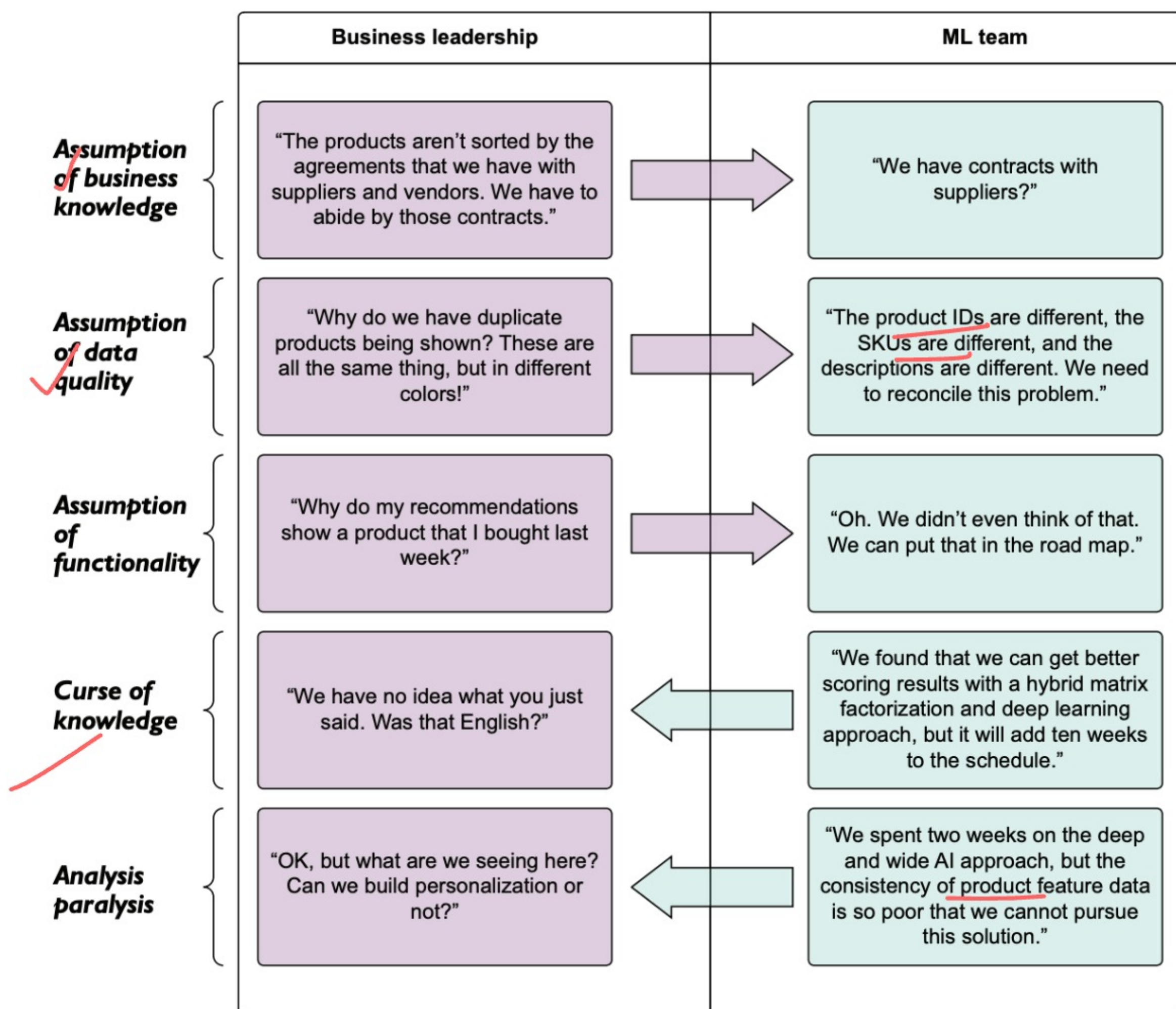
However, it's one of the most effective and important questions, as it opens a discussion into the true motivations for why people want the project to be built.

- 1) Is it to increase sales?
- 2) Is it to make our external customers happier?
- 3) Or is it to keep people browsing on the site for longer durations?

Figure below sums up the planning process for the MVP. With extremely vague requirements, a complete lack of thorough communication about expectations for



Source:- Machine learning in action, author:- Ben Wilson



Source:- Machine learning in action, author:- Ben Wilson

Figure above shows the questions and the underlying root causes of the present misunderstandings.

3.2.1) Assumption of Business Knowledge

Case Study: E-Shop's Personalization Failure



Context

E-commerce company built a personalized recommendation system



What Went Wrong

ML model violated business rules due to assumed knowledge



Why It Happened

No SME involved; relied on assumptions



How It Could've Been Avoided

Engage stakeholders to clarify requirements

Engage stakeholders to clarify requirements

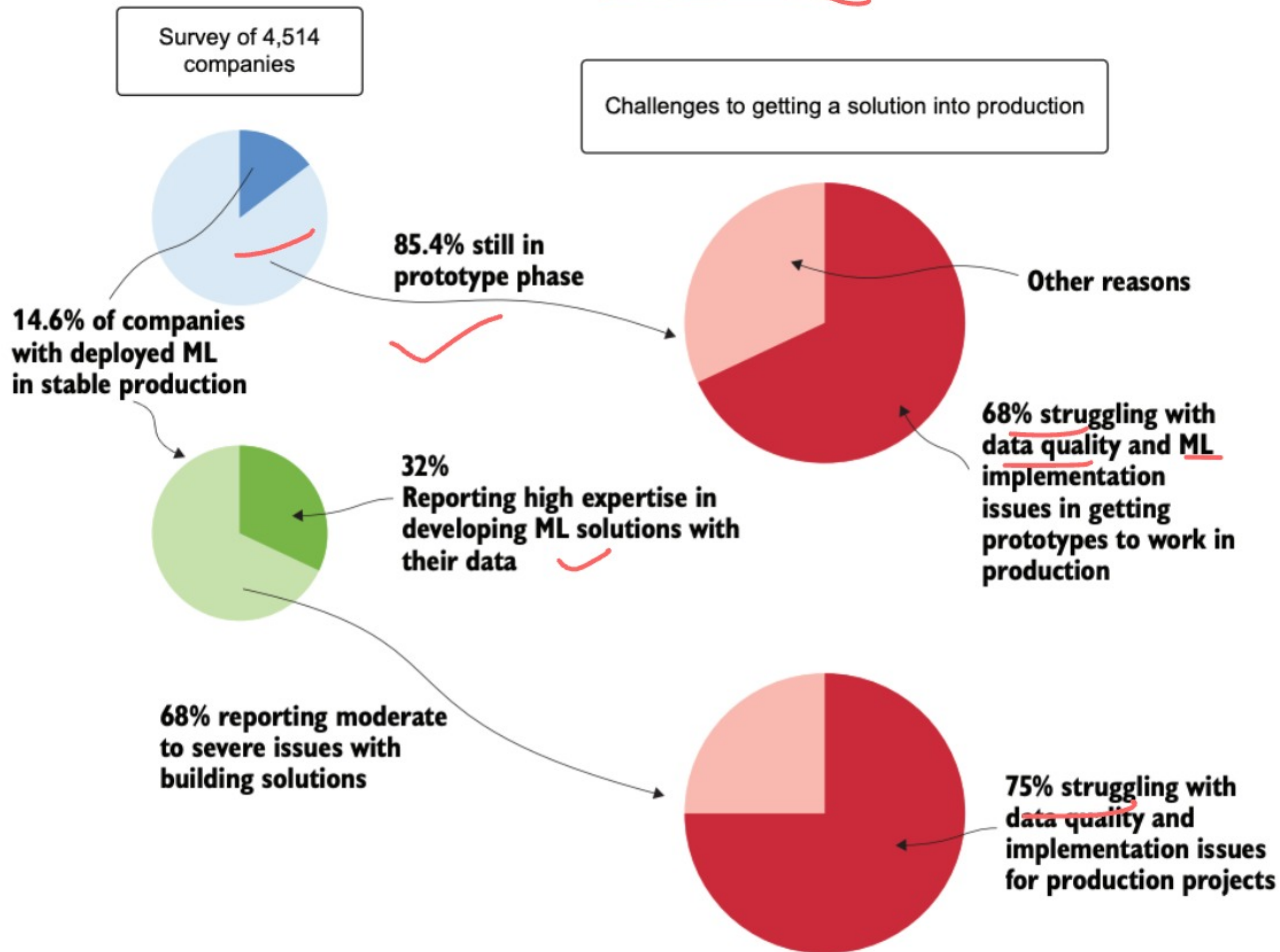
Always begin with clarity, aligning on business goals, assumptions, and stakeholder inputs early ensures machine learning projects remain focused, feasible, and impactful.

3.2.2) Assumption of Data Quality

Misalignment of expectations: Duplicate product listings in the prototype were not anticipated due to a lack of early discussion between business and ML teams, eroding confidence in the ML output despite technical correctness.

Assumed data quality: The ML team relied on the DE team's assertion that the data was clean, overlooking deeper validation and assumptions around product listings and categorization.

Industry reports on the impact of data quality and cleanliness on AI (ML) adoption



Sources:

1. 2020 Deloitte Insights survey on State of AI in the Enterprise
2. 2020 IBM Survey Results for AI Adoption and Challenges "From Roadblock to Scale: The Global Sprint Towards AI"

The impact of data quality issues on companies engaging in ML project work. Data quality issues are common, and as such, should always be vetted during the early stages of project work.

Lesson learned: Effective ML planning must include data validation, collaboration with SMEs, and early questioning of assumptions like "Are all products logged uniformly?", to avoid flawed models and wasted effort.

3.2.3) Assumption of Functionality



Assuming functionality without clarifying user expectations can silently undermine an ML project. What seems like a minor oversight to the technical team, like recommending a recently purchased item, can seriously affect user trust. Addressing such gaps early through structured feedback and clear scoping ensures the solution truly aligns with end-user needs.

3.2.4) Curse of Knowledge

- 1) Avoid overwhelming non-technical stakeholders with complex jargon as it hinders understanding and engagement.
- 2) Communicate technical efforts in abstract terms, such as saying multiple approaches were tested with varying impacts on time and performance.
- 3) Let your audience guide how technical the discussion should get, instead of assuming their level of expertise.

3.2.5) ANALYSIS PARALYSIS



Example: Shoe recommendation system

The ML team develops the latest model for generating personalized recommendations



A simple approach works too

Consider using a simple rule-based system



For example, excluding the most recently purchased items

In the world of machine learning, overanalyzing every possible model, metric, or parameter can stall progress more than poor modeling ever will. Analysis paralysis creeps in when teams focus too much on achieving perfection instead of delivering value.

The goal should not always be to build the *best* model, but to build a *useful* one, one that meets business needs, integrates well into workflows, and can be iteratively improved.

The key is to scope smartly, prioritize simplicity, and stay grounded in real-world requirements. Progress comes from delivering a solution that works today, not obsessing over one that might be marginally better tomorrow.

We will continue this chapter in the Next Lecture, We will focus on some other aspects of Planning and then we will understand more about scoping.

THANK
YOU

